#3.

2変量確率変数 (X, Y)

Prob. mass function:

① 同時 PMF $\rightarrow P(X=x_i, Y=y_j)$

② 周辺 PMF.

$\rightarrow \sum_{y \in \mathcal{Y}} P(X=x_i, Y=y) = P(X=x_i)$

$\sum_{x \in \mathcal{X}} P(X=x, Y=y_j) = P(Y=y_j)$ (R.V.) X

③ 条件付き PMF.

$P(X=x_i \mid Y=y_j)$

$= \dfrac{P(X=x_i, Y=y_j)}{P(Y=y_j)} = \dfrac{\text{同時 PMF}}{\text{周辺 PMF}}$

$\sum_{x_i \in \mathcal{X}} P(X=x_i \mid Y=y_j) = 1$

$= \dfrac{P(x_1 \mid y_j)}{P(y_j)} + \dfrac{P(x_2 \mid y_j)}{P(y_j)} + \cdots \dfrac{P(x_i \mid y_j)}{P(y_j)}$

$= \dfrac{P(y_j)}{P(y_j)} = 1$

Y ③ $P(X=x_i \mid Y=y_j)$ $X=x_1$ の周辺 PMF.

② Marginal total.

	y_1	y_2	$\cdots$	y_j	$\cdots$	
(実現値) x_1	$P(x_1, y_1)$	$P(x_1, y_2)$				$P(x_1, y_1) + P(x_1, y_2) + \cdots P(x_1, y_j) + \cdots = P(X=x_1)$
x_2	$P(x_2, y_1)$	$P(x_2, y_2)$				
$\vdots$ x_i				① $P(x_i, y_j)$		$P(X=x_i)$
$\vdots$						
Marginal total ②	$P(y_1, x_1) + P(y_1, x_2) + \cdots + P(y_1, x_i) + \cdots = P(Y=y_1)$			$P(Y=y_j)$		

$Y=y_1$ の周辺 PMF

$P(Y=y_j \mid X=x_i)$

$$E\{E[g(X,Y)|X]\}$$

$X \in \mathcal{X} = \{x_1, x_2, \dots\}$

$Y \in \mathcal{Y} = \{y_1, y_2, \dots\}$

固定した条件.

(条件付き)

変数 → Yの期待値を求める

$$\boxed{E[g(X,Y)|X]} = \sum_{y \in \mathcal{Y}} g(x,y) \cdot \frac{P(X=x, Y=y)}{P(X=x)}$$

$g(X=x, Y=y_2) \to g(x)$

$$= g(x\ y_1)\frac{P(x,y_1)}{P(x)} + \boxed{g(x,y_2)\cdot\frac{P(x,y_2)}{P(x)}} + \cdots + g(x,y_j)\cdot\frac{P(x,y_j)}{P(x)} + \cdots$$

$$E\{\square\} = E\left[g(x\ y_1)\cdot\frac{P(x\ y_1)}{P(x)}\right] + E\left[g(x\ y_2)\frac{P(x\ y_2)}{P(x)}\right] + \cdots + \cdots$$

変数　Y=y₁:固定した実現値

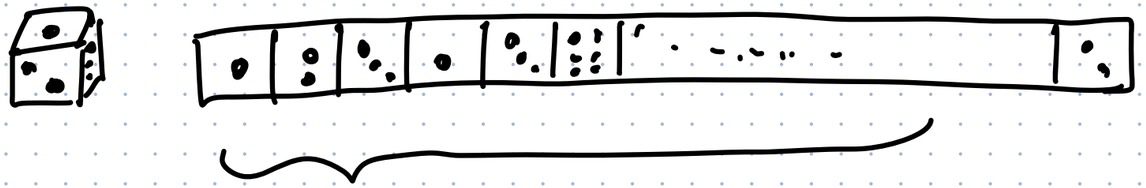
$$= \overset{①}{\sum_{x=0}^{\infty}} g(x\ y_1)\cdot\frac{P(x\ y_1)}{P(x)}\cdot P(x) + \overset{②}{\sum_{x=0}^{\infty}} g(x\ y_2)\frac{P(x\ y_2)}{P(x)}\cdot P(x) + \cdots\cdots + \overset{ⓘ}{\sum_{x=0}^{\infty}}$$

$$\begin{aligned}
&① \quad g(x_1\ y_1)\cdot\frac{P(x_1\ y_1)}{P(x_1)}\cdot P(x_1) + g(x_2\ y_1)\cdot\frac{P(x_2\ y_1)}{P(x_2)}\cdot P(x_2) + \cdots \\
&+ \\
&② \quad g(x_1\ y_2)\cdot\frac{P(x_1\ y_2)}{P(x_1)}P(x_1) + g(x_2\ y_2)\cdot\frac{P(x_2\ y_2)}{P(x_2)}\cdot P(x_2) + \cdots \\
&+ \\
&\vdots \\
&+ \\
&ⓘ \quad g(x_1, y_j)\cdot\frac{P(x_1, y_j)}{P(x_1)}P(x_1) + g(x_2, y_j)\cdot\frac{P(x_2, y_j)}{P(x_2)}\cdot P(x_2) + \cdots
\end{aligned}$$

(略記)

$$= \sum_{x=0}^{\infty}\sum_{y=0}^{\infty} g(x,y)\cdot P(x,y) = E[g(X,Y)]$$

Multinomial Distribution 多項分布



n回投げる

X_1: ⚀が出る回数 ⇒ #⚀　　$P(⚀出)=p_1$

X_2: #⚁　　$P(⚁出)=p_2$

X_3: #⚂

⋮

X_6: #⚅　　$P(⚅)=p_6$

$p_1+p_2+\cdots+p_6=1$

$X_1=x_1$ 回

$X_2=x_2$ 回

$X_3=x_3$ 回

⋮

$X_6=x_6$ 回

$x_1+x_2+\cdots+x_6=n$ 回

$$P(X_1=x_1, X_2=x_2, \ldots, X_6=x_6)=\binom{n}{x_1}\binom{n-x_1}{x_2}\binom{n-x_1-x_2}{x_3}\cdots p_1^{x_1}p_2^{x_2}\cdots p_6^{x_6}$$

$$=\frac{n!}{x_1!\,x_2!\cdots x_6!}p_1^{x_1}\cdots p_6^{x_6}$$ (多項分布)

Ⓣ
Ⓗ

n回

X_1: Tが出る回数 ⇒ #T　　$P(T)=p_1 \Rightarrow P(H)=1-p_1$
$=p_2$

X_2: #H $=n-X_1$

$$P(X_1=x_1, X_2=n-x_1)=\binom{n}{x_1}\cdot p_1^{x_1}(1-p_1)^{n-x_1}$$

(二項分布)